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CMSE, Michigan State University
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CURRICULUM VITAE

EDUCATION

Brown University Division of Applied Math Providence, RI, USA	Ph.D. , 2018 - 2023 <i>Thesis Title: Applications of FEM with Macro-Elements: Discrete Elasticity Sequences and Convergence of Lagrange Elements for a Maxwell Eigenvalue Problem</i> <i>Advisor: Dr. Johnny Guzmán, Professor</i>
Beijing Normal University School of Mathematical Sciences Beijing, China	B.S. , 2014 - 2018 <i>Thesis Title: The Green's Function for the Helmholtz Equation on Special Domains</i> <i>Advisor: Dr. Haigang Li, Professor</i>

PROFESSIONAL EXPERIENCE

2023.8 -	Postdoctoral Associate. Center of Hierarchical and Robust Modeling for Non-Equilibrium Transport (CHaRMNET), Michigan State University.
2018.9 - 2023.5	Graduate Student Researcher. Division of Applied Mathematics, Brown University.
2016.5 - 2017.5	Team leader. National Undergraduate Training Program for Innovation and Entrepreneurship, <i>Tracking Targets by Particle Filters Based on Color Histogram</i> , Beijing Normal University.

RESEARCH INTERESTS

- High order numerical methods for fluid and kinetic equations
- WENO/ENO methods, finite element methods and computational fluid dynamics
- Finite Element Exterior Calculus and Spline theories
- Low rank (tensor) approximations
- Method of line transpose (Kernel-based operators)

HONORS & AWARDS

2023	The Stella Dafermos Award. The prize goes to one or more outstanding graduate students in honor and memory of Stella Dafermos, Brown University.
2020.6	Graduate School 2020 Summer Fellowships , Brown University.
2018.6	Outstanding Graduates of the University , Beijing Normal University.
2018.6	Outstanding Thesis Award of the Department , Beijing Normal University.
2015/2016/2017	The First Prize Scholarship , Beijing Normal University.

Published or Accepted

- **Gong, S.**, Guzmán, J. & Neilan, M. A Note on the Shape Regularity of Worsey–Farin Splits. *Journal of Scientific Computing*, 2023.
- Boffi, D., **Gong, S.**, Guzmán, J. & Neilan, M. Convergence of Lagrange finite element methods for Maxwell eigenvalue problem in 3D. *IMA Journal of Numerical Analysis*, 2023.
- **Gong, S.**, Gopalakrishnan, J., Guzmán, J. & Neilan, M. Discrete elasticity exact sequences on Worsey-Farin splits. *ESAIM: M2AN*, 2023.
- Christlieb, A., **Gong, S.**, & Yang, H. Boundary corrections for kernel approximation to differential operators. *Journal of Scientific Computing*, 2025.
- Padilla-Gomez, FA., **Gong, S.**, Murillo, MS., Graziani, FR., Christlieb, A. Quantum Kinetic Modeling of KEEN waves in a Warm-Dense Regime. *Physics of Plasmas*, 2026.

Accepted Preprints

- Christlieb, A., **Gong, S.**, Qiu, J., & Zheng, N. A Sampling-Based Adaptive Rank Approach to the Wigner-Poisson System. Submitted to *SIAM Journal on Scientific Computing*.

Submitted Manuscripts

- Christlieb, A., Chacón, L., & **Gong, S.**. An Energy-Conserving Unstaggered Electromagnetic-Potential Particle-in-Cell Method, Part I: Non-relativistic Generalized-Momentum Formulation. Submitted to *SIAM Journal on Scientific Computing*.
- Christlieb, A., **Gong, S.**, Qiu, J., & Zheng, N. A Structure-preserving Adaptive-Rank Approach to the High-Dimensional Wigner-Poisson System. Submitted to *SIAM Journal on Scientific Computing*.
- Christlieb, A., **Gong, S.**, Padilla-Gomez, FA., & Qiu, J. A Conservative Adaptive Rank Method for the Wigner-Poisson System. Submitted to *Journal of Scientific Computing*.

PRESENTATIONS (SELECTED)

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| 2026.7 | <i>A Structure-preserving Adaptive-Rank Approach to the High-Dimensional Wigner-Poisson System</i> , 2026 SIAM Annual Meeting. |
| 2025.11 | <i>A Sampling-Based Adaptive Rank Approach to the Wigner-Poisson System</i> , 2025 SIAM NNP. |
| 2025.8 | <i>A Kernel-Based Time-Stepping Framework for Stiff PDEs: Revisiting MOLT with SSP-RK</i> , Innovative and Efficient Strategies for Stiff Differential Equations workshop @ ICERM. |
| 2025.7 | <i>A Sampling-Based Adaptive Rank Approach to the Wigner-Poisson System</i> , The 15th International Conference on Spectral and High Order Methods (ICOSAHOM 2025). |
| 2024.7 | <i>High order unconditionally strong stability preserving (SSP) implicit two-derivative Runge–Kutta schemes and their extensions</i> , 2024 SIAM Annual Meeting (AN24). |
| 2023.2 | <i>Convergence of Lagrange FEM for Maxwell Eigenvalue Problem in 3D</i> , Pitt AWM Student Seminar, University of Pittsburgh. |
| 2022.12 | <i>Convergence of Lagrange FEM for Maxwell Eigenvalue Problem in 3D</i> , Math Colloquium, Michigan Technological University. |
| 2022.10 | <i>A Discrete Elasticity Exact Sequence on Worsey-Farin Splits</i> , Finite Element Circus, Carnegie Mellon University. |
| 2022.4 | <i>Convergence of Lagrange FEM for Maxwell Eigenvalue Problem in 3D</i> , Finite Element Circus, University of Florida. |

PROFESSIONAL SERVICE

- *Results in applied mathematics.* - Reviewer (2024)
- *Computers & mathematics with applications.* - Reviewer (2024, 2 manuscripts)
- *Computer Aided Design.* - Reviewer (2025)
- *Communications on Applied Mathematics and Computation.* - Reviewer (2025)

TEACHING AND MENTORING EXPERIENCE

Teaching activities

Fall 2025/ Spring 2024	MTH 132 Calculus I, <i>Instructor</i> , Michigan State University.
Spring 2023	APMA 1160 An Introduction to Numerical Optimization, <i>teaching assistant</i> , Brown University.
Fall 2022	APMA 1690 Computational Probability and Statistics, <i>teaching assistant</i> , Brown University.
Spring 2020	APMA 340 Methods of Applied Mathematics II, <i>head teaching assistant</i> , Brown University.
Fall 2019	APMA 330 Methods of Applied Mathematics I, <i>teaching assistant</i> , Brown University.
Summer 2019/2022	Summer Review Course for Analysis, <i>teaching assistant</i> , Brown University.

Mentoring activities

- Francisco Alejandro Padilla Gomez, PhD Student (MSU). Co-mentoring on the Wigner-Poisson solver for the KEEN waves and Conservative Wigner-Poisson solver with Prof. Andrew Christlieb.
- Doyoung Wang, Undergraduate student (MSU). Instruction on the Schrödinger equations, Wigner-Poisson equations and the implementing of Wigner-Poisson solvers.

OTHER SKILLS AND EXPERIENCE

Languages: Mandarin (native) and English (fluent)

Programming Skills: Matlab, FEniCS, Python, LaTeX (significant experience) and C/C++, Mathematica, Julia (some experience).

Undergraduate minor: Psychology.